



Do Negative Brand Reviews Impact All Users or Specific Customer Segments Only?

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Section B – Group 3



1. Introduction

1.1 The Dataset

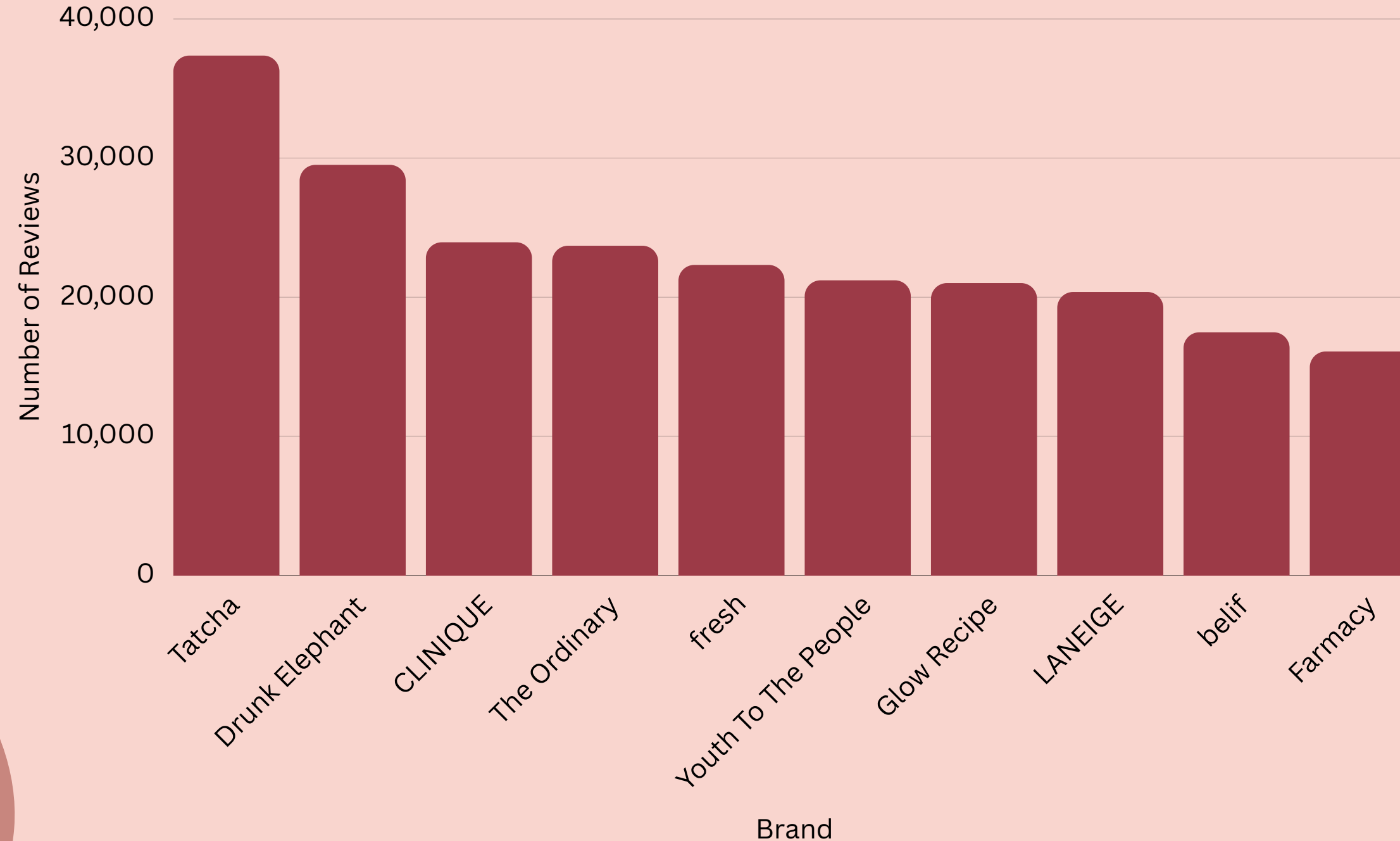
- **519,409** Sephora product reviews representative sample
- Product metadata, review text and reviewer skin attributes
- Sentiment labels:
 - 4-5 = Positive (82%)
 - 3 = Neutral (7.5%)
 - 1-3 = Negative (10.4%)
- TF-IDF feature cap: 1,000 terms
- Train / test split: 80% / 20% stratified
- Large corpus but constraints:
 - Label construction strategy
 - Class imbalance
- **Problem** – brands struggle to identify:
 - Impact of negative reviews
 - Most problematic product types
 - Demographic groups most at risk



1.2 Why Sephora?



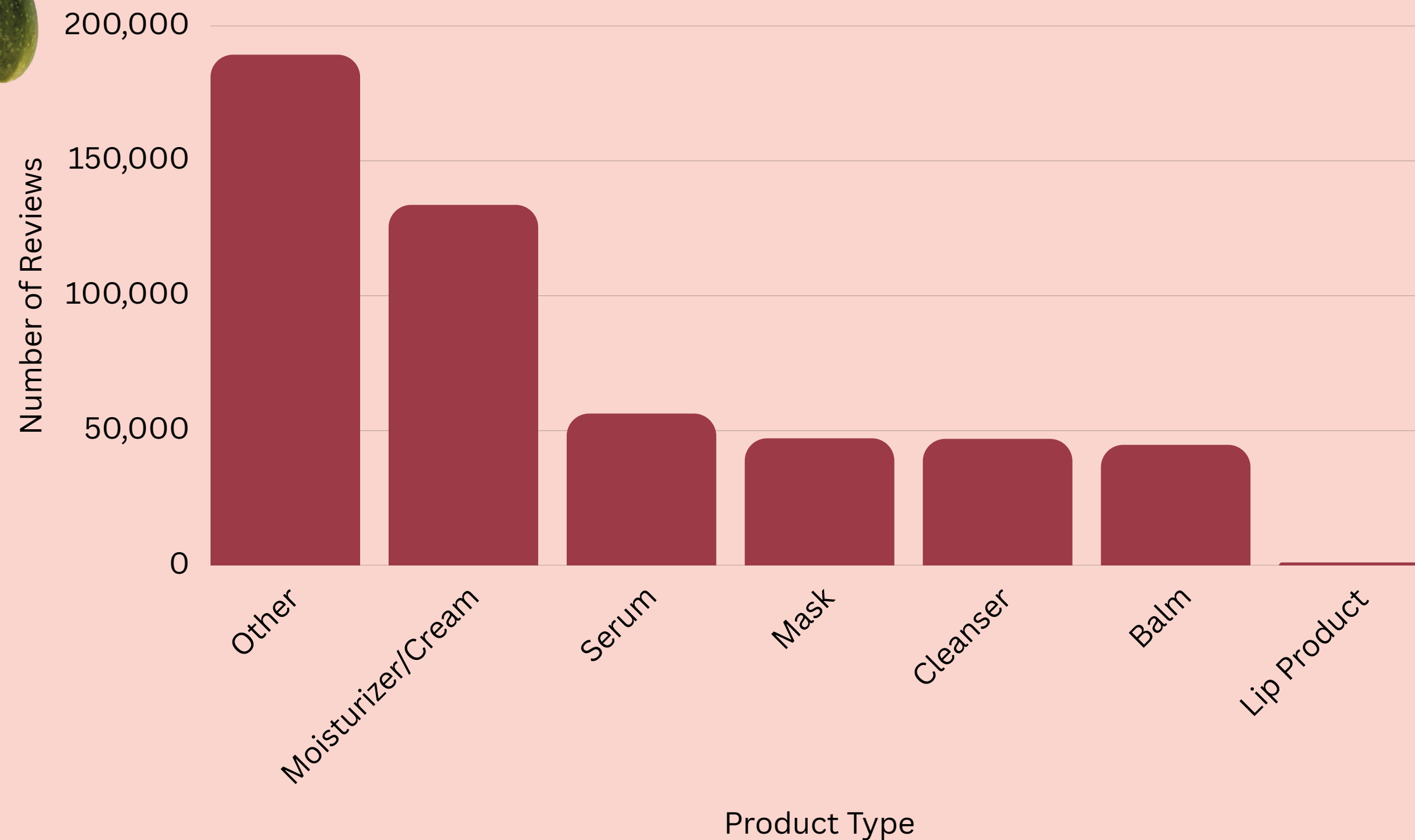
- Leading luxury global beauty retailer
- Growing while competitors see downturn
 - Resilient through global uncertainty and inflation
- Carrara Advisory: Sephora is a leader in European premium beauty:
 - €4 million productivity per store
 - High foot traffic, high spend, exclusive brands, loyalty programmes
- Despite LVMH hit in 2025 fiscal year, Sephora remained profitable
 - Gained market share in several countries
 - Opened c. 100 stores in the same year
- Brand exclusivity: Korean skincare brands
- Sephora.fr accounts for c. 15% of online beauty spending
- High-quality in-house brand, a top seller





2.1 Top Brands and Products

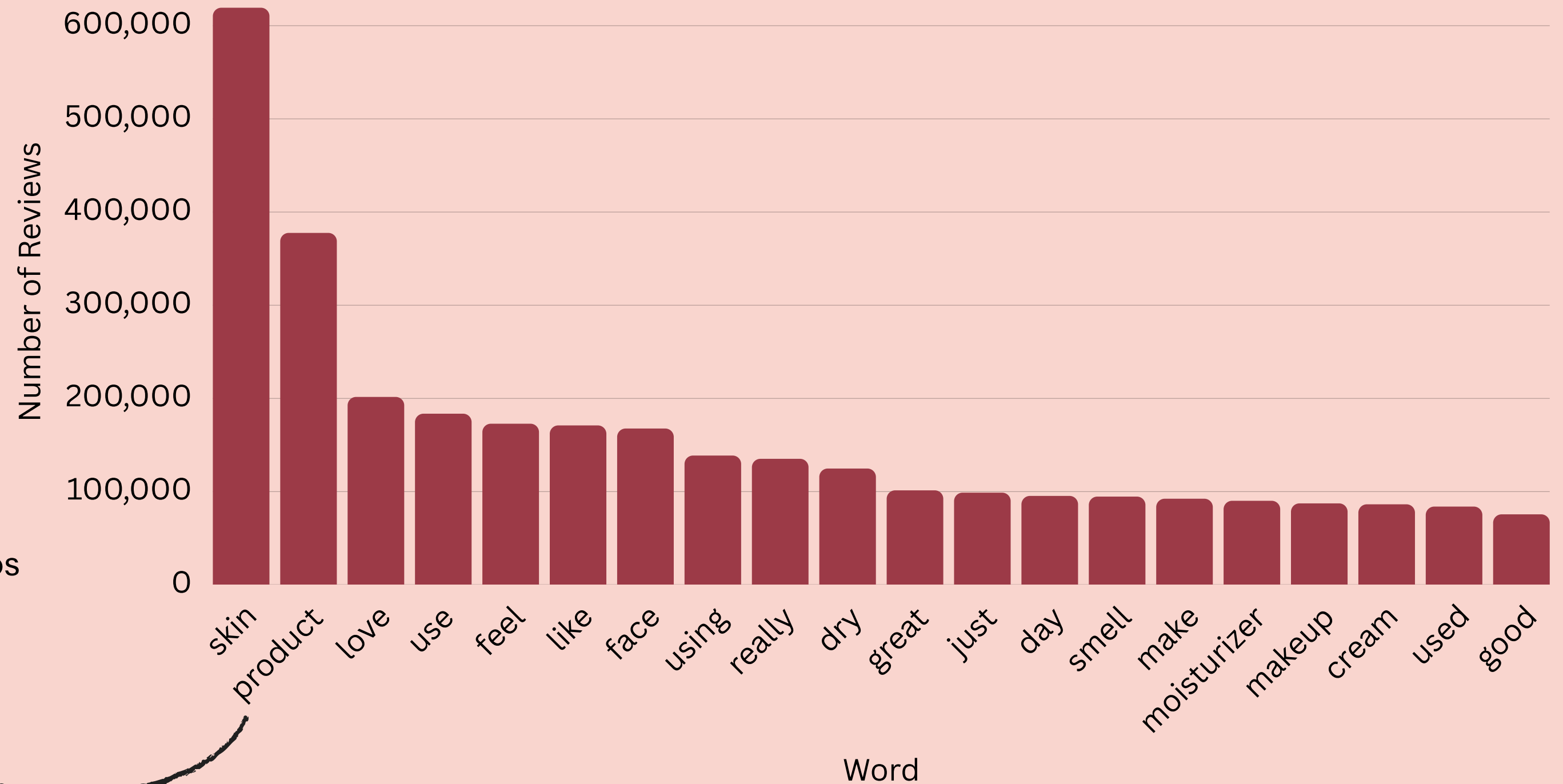
Distribution of Product Type



- **Other category dominates:** mapping function captures limited set of product keywords
- **Moisturizer/Cream** = strongest recognised category
- Heavily **skincare** oriented

2.2 Review Text Patterns: Frequency

Top 20 Most Frequent Words

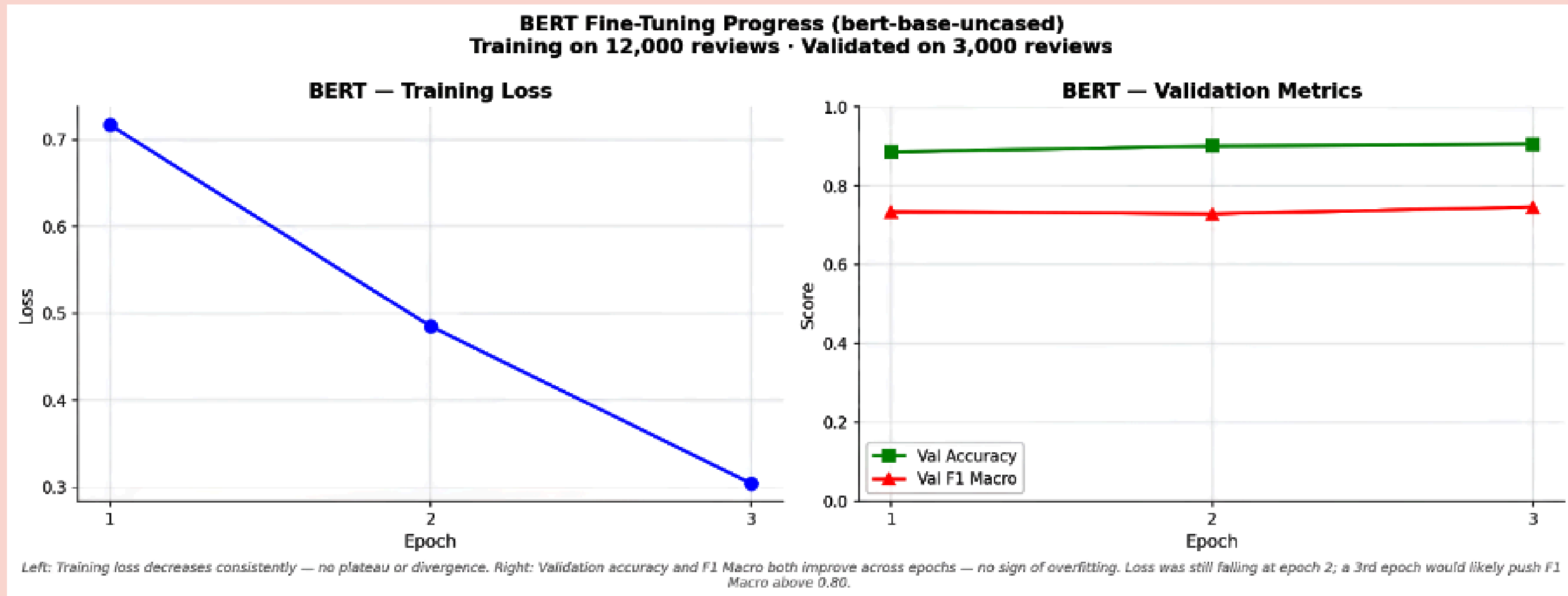


Preprocessing pipeline
lowercases text, removes
non-letter characters, drops
English stop words and
lemmatizes the remaining
tokens

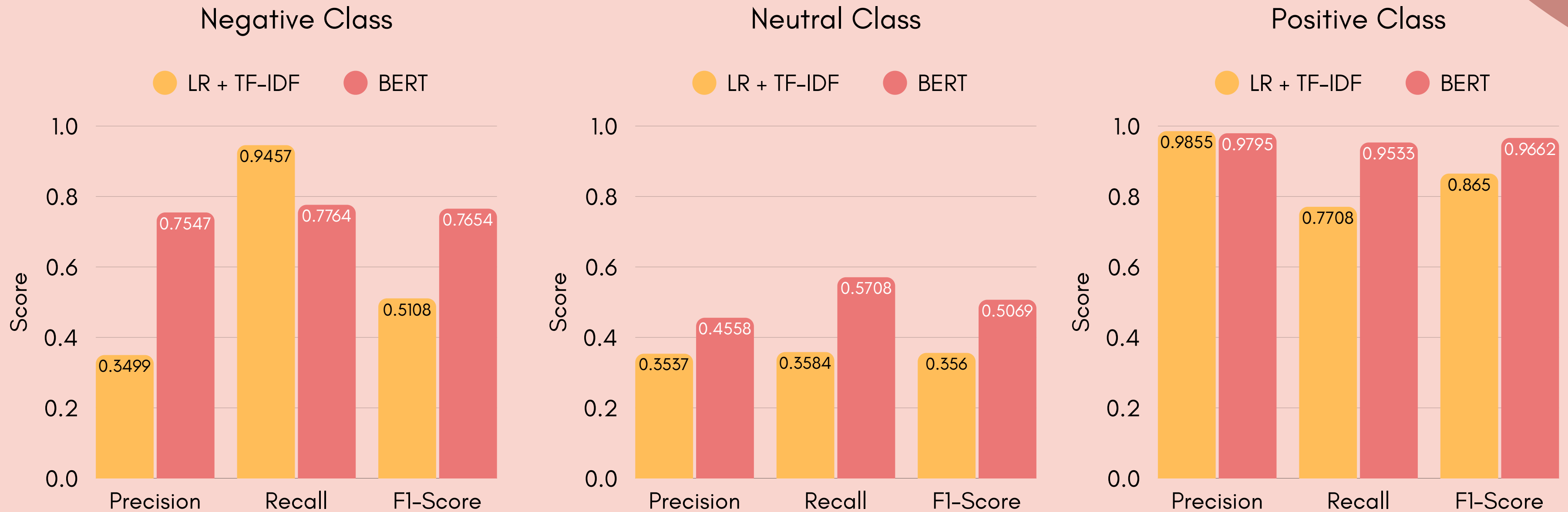


3. Sentiment Models

3.1 BERT: Results



3.2 Comparing LR with BERT



- Neg. F1 = 0.76 (+26pp), Precision = 0.75 (+41pp vs. LR)
- BERT = fewer false alarms and more actionable

bert-base-uncased · 110M params · 3 epochs · 12k training / 3k validation reviews.

4. LDA and Segmentation

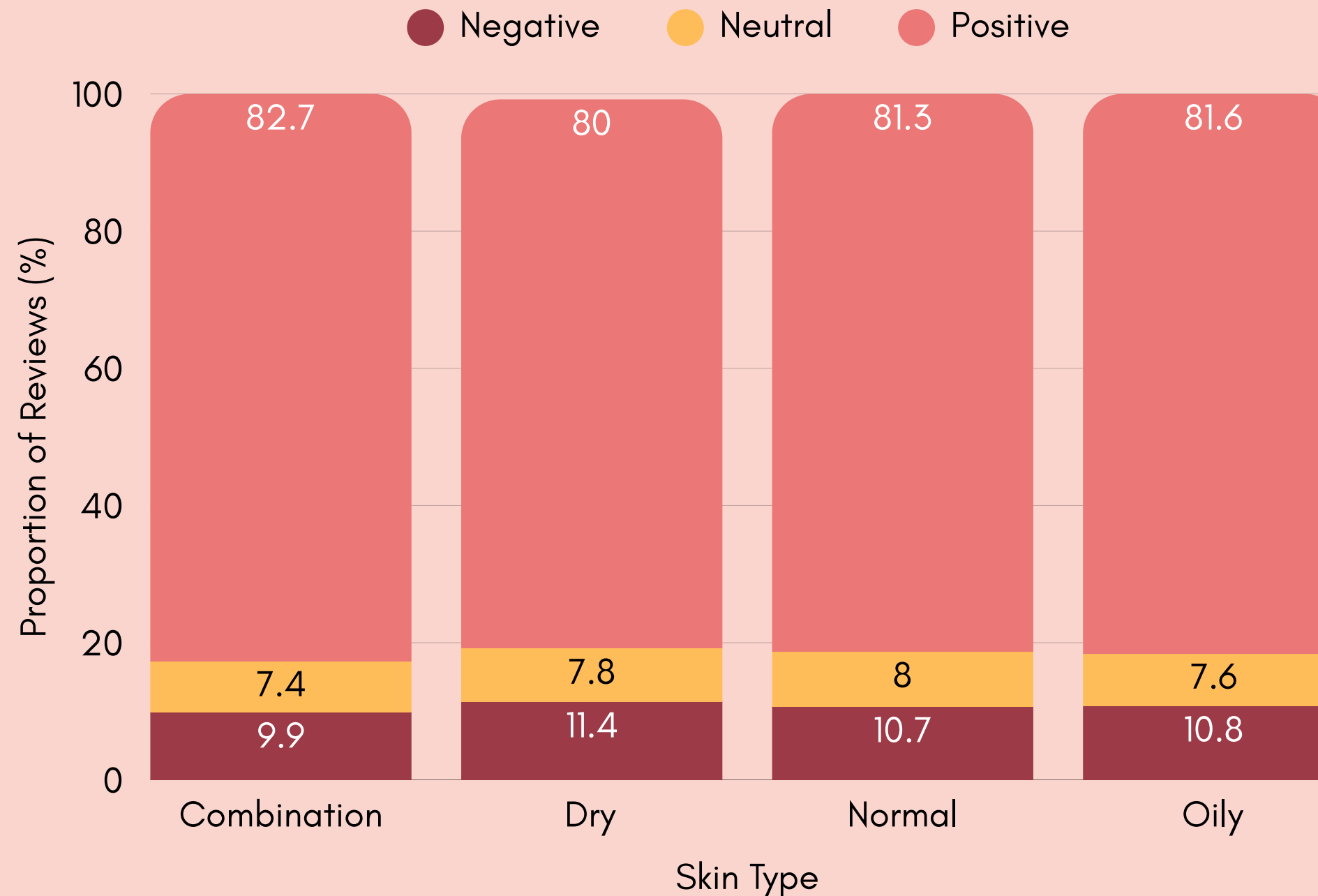
7 complaint themes from LDA, which are **not random** and cluster:

- 1.Lip & Makeup Removal
- 2.Acne & Time Effects
- 3.Product Efficacy
- 4.Eye Products
- 5.Sensitive Skin
- 6.Dryness & Moisturizer
- 7.Texture & Smell

Disclaimer: LDA Topic chart is on slide number 20

4.1 Demographic Segmentation

Sentiment Distribution by Skin Type
Normalised Proportions

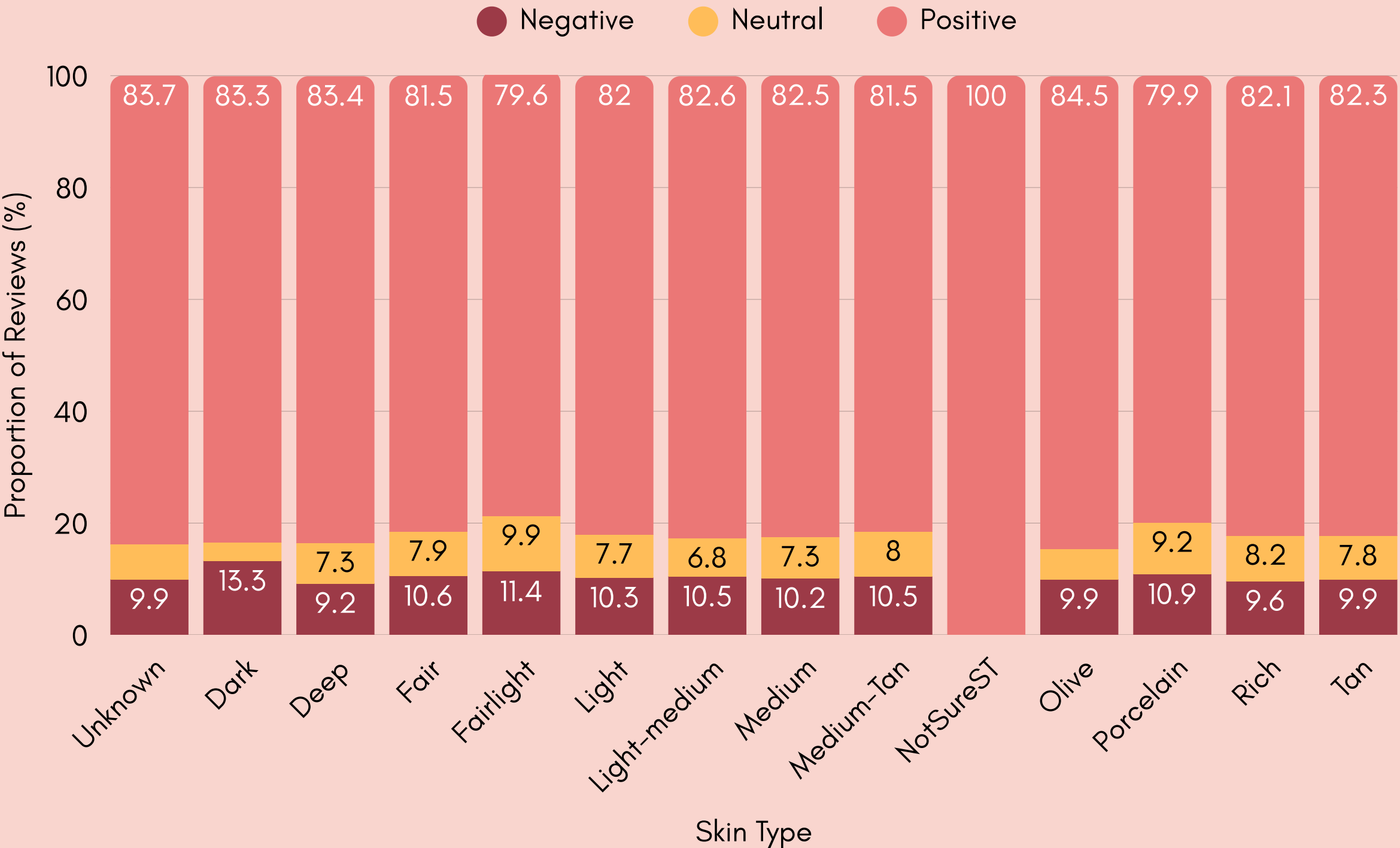


All 4 types show similar negative review rates (c. 10-11%)



4.1 Demographic Segmentation

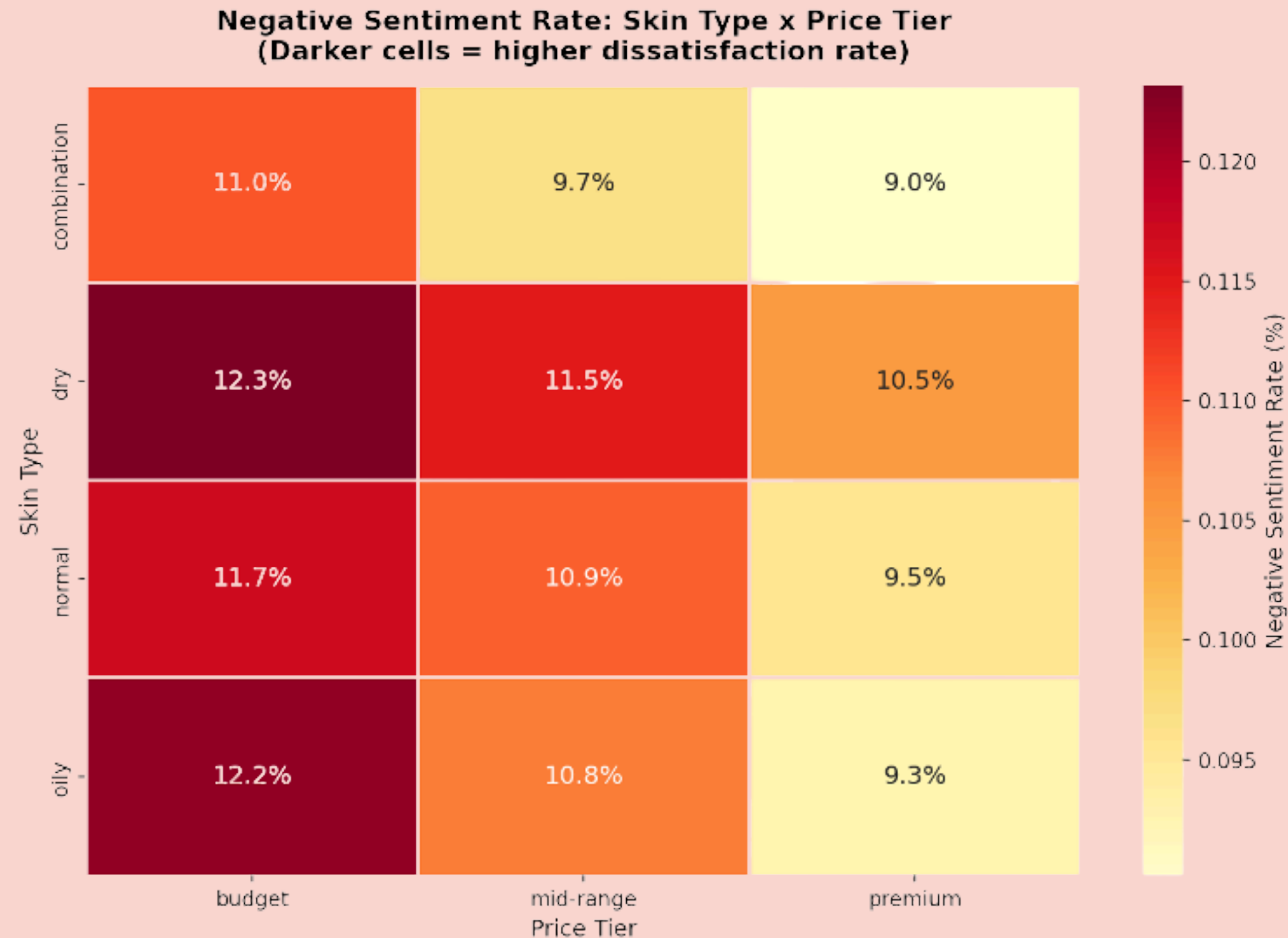
Sentiment Distribution by Skin Type
Normalised Proportions



- Dark/rich tones: c.14–15% negative vs. c.9% for fair and porcelain
- Dissatisfaction is **segment specific**:
 - Darker tones face a 67% higher negative rate
 - Not uniform across all customers
 - Suggests an inclusivity issue or shade mismatch problem

4.2 Price-Tier Analysis

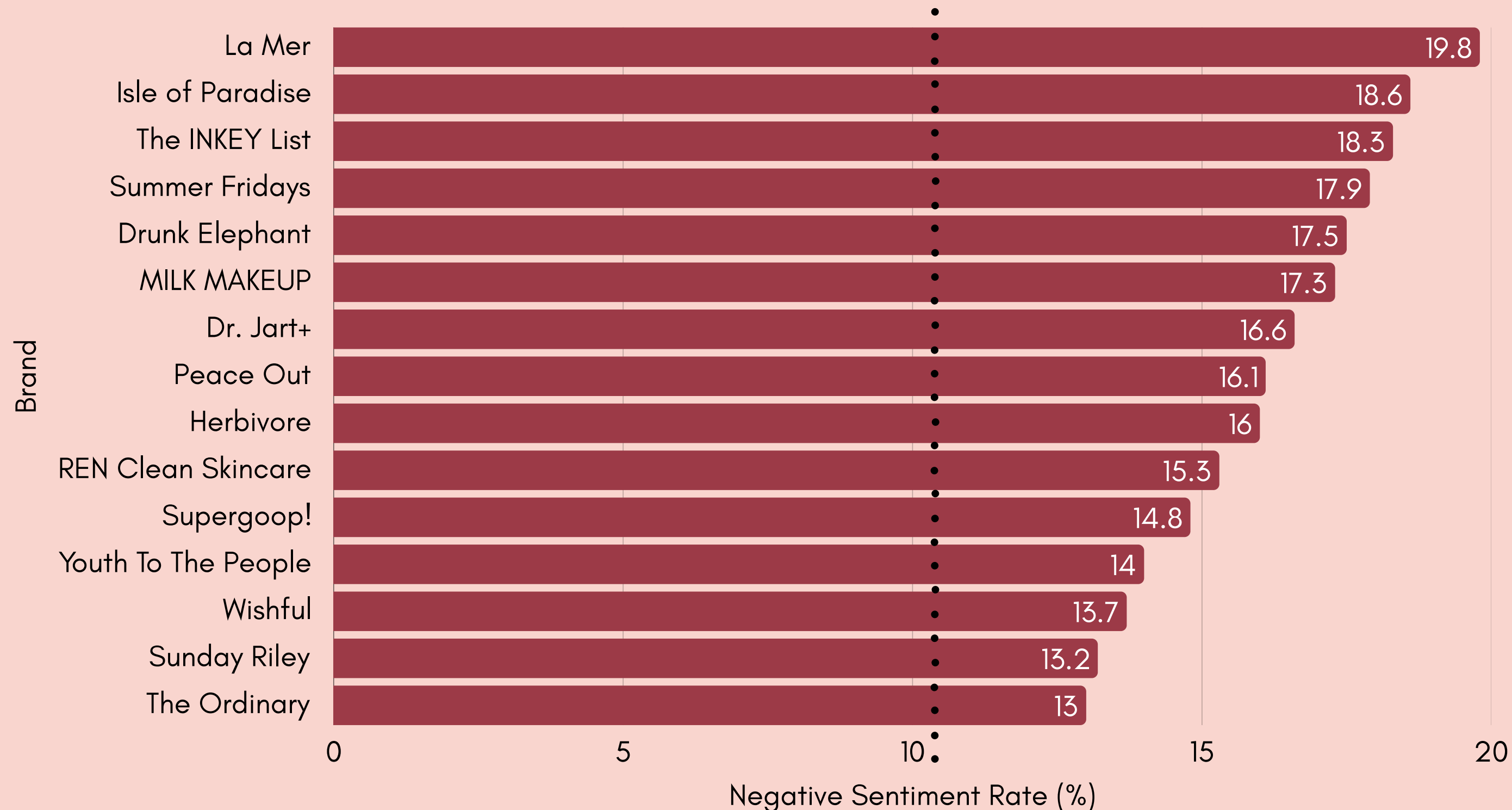
- Cross-tabulation:
negative sentiment rate
(%) x price tier cell
- Negative rate: skin type
x price tier
- **Worst segment:**
Budget + Dry skin
(12.3%)
- **Best combination:**
Premium + Combination
skin (9%)



4.3 Brand Sentiment

Top 15 Brands by Negative Sentiment Rate

Brands with ≥ 30 reviews. Dashed line = average (10.4%)



- Skin type alone does not differentiate dissatisfaction; it **must cross** with price tier
- E.g. La Mer = 19.8% negative
- Problem is not product quality but **value perception**
- Pricing strategy matters as much as **product quality**, and customer expectations increase with **price**



5. Insights & Recommendations

1) **Reformulate for dry and sensitive skin**

- Highest dissatisfaction (c. 12.3%) = dry and sensitive skin + budget segment
- Product-skin mismatch
 - Reformulate for greater suitability for skin types e.g. fragrance-free, hypoallergenic
 - Clear labelling of skin compatibility
 - Satisfaction and brand trust

2) **Expand shade range for deeper tones**

- Darker or richer skin tones = higher negative review rate (c. 14-25%) = 67% higher relative dissatisfaction
- LDA: gap in product inclusivity
 - Expand and tailor shade offerings
 - Design and test products across diverse skin tones
 - Brand inclusivity and market reach



3) Address value-for-money perception

- La Mer leads at c. 20% negative reviews despite premium pricing
- LDA: 'money', 'price', 'not worth' = complaint drivers
 - Sample or trial sizes
 - Improve transparency around product benefits and timelines
 - Shift marketing towards outcome-based messaging to align with expectations

4) Deploy BERT for real-time monitoring

- Improves sentiment detection (0.78) and precision
- Reduces false alarms and enables faster, reliable decision-making
- Real-time monitoring for incoming reviews:
 - Negative sentiment detection
 - Recurring issues
- Use alongside logistic regression for greater interpretability





Thank You
Any Questions?

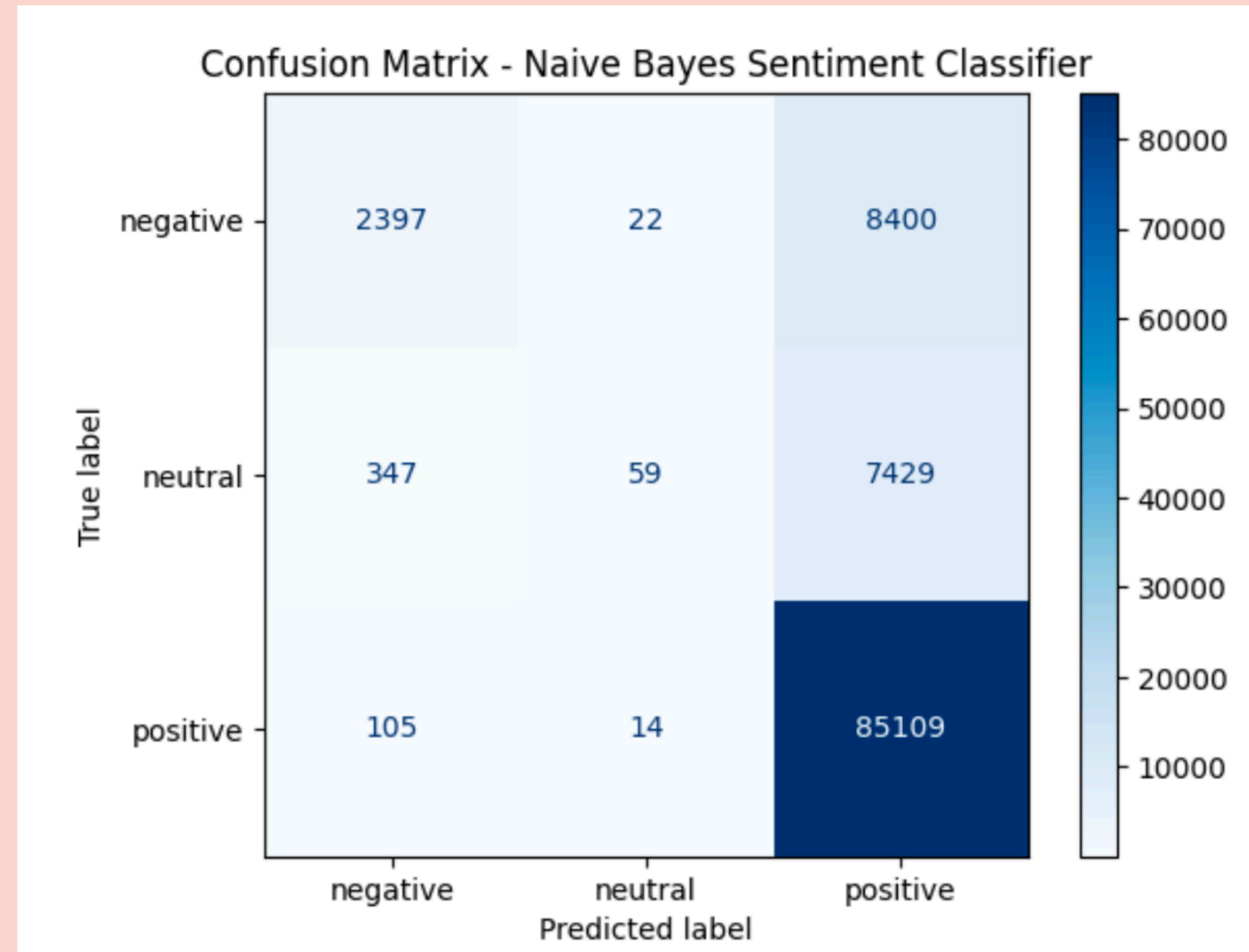


Appendix

1. Baseline Sentiment Model

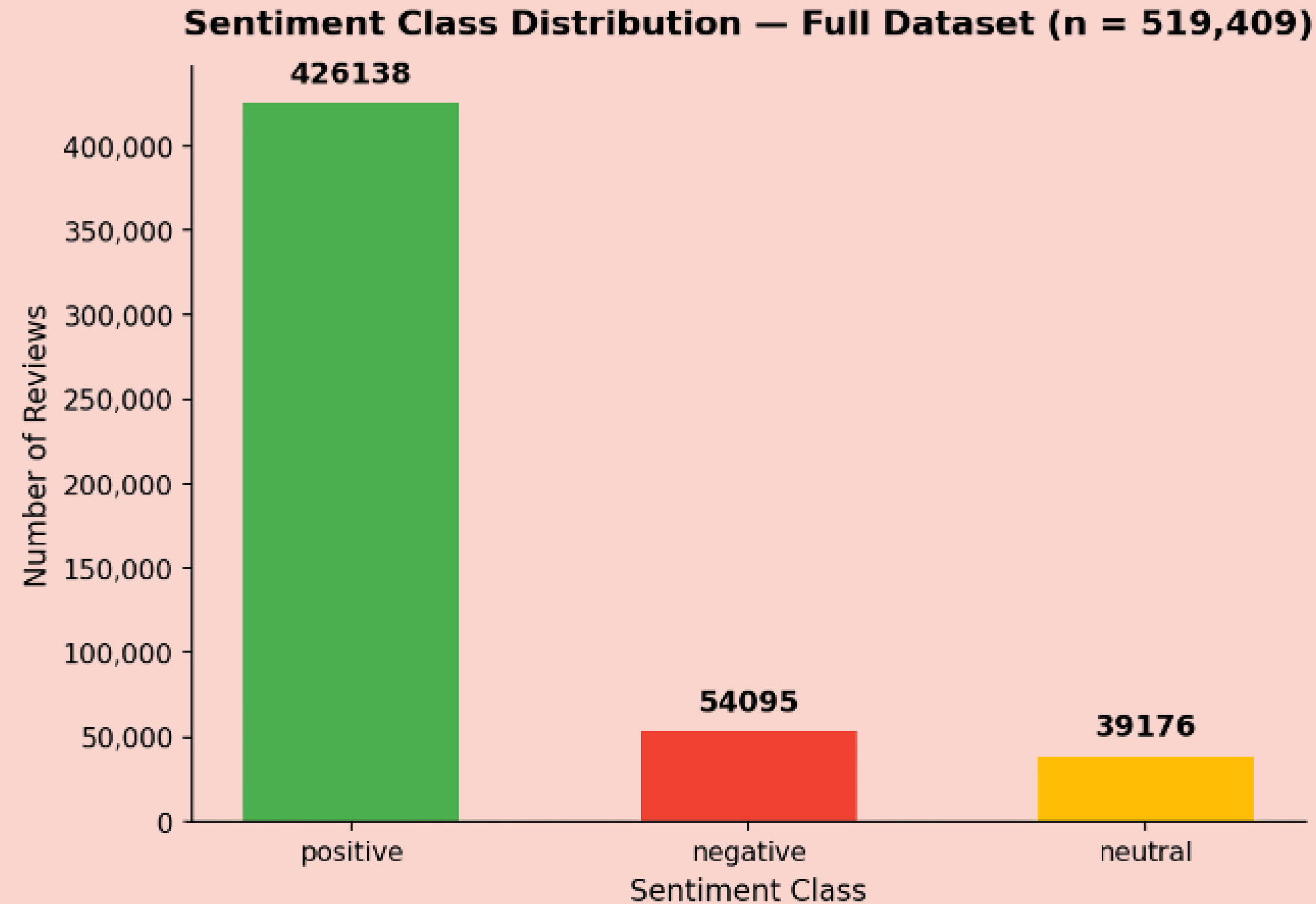
Class	Precision	Recall F1-score
negative	0.84	0.22 0.35
neutral	0.62	0.01 0.01
positive	0.84	1.00 0.91

- Overall accuracy: 0.8429
- Test set size: 103,882
- Model is a useful baseline but it mainly succeeds by identifying positive reviews
- Neutral reviews are almost never recovered
- Recall for negative reviews is weak
- In practice, classifier is not yet reliable for balanced customer sentiment monitoring or escalation workflows



Appendix

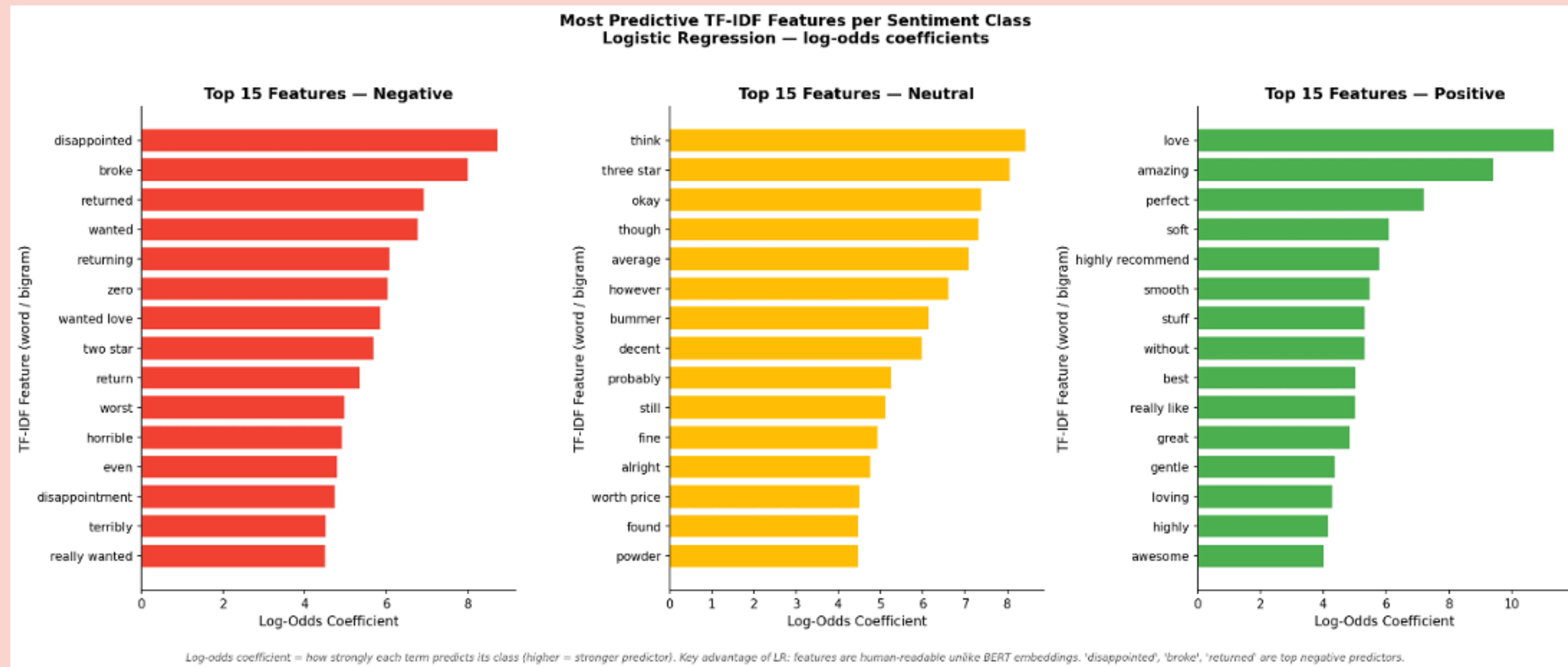
2. Sentiment Distribution of the dataset



Labels derived from star ratings: 4-5 stars = Positive | 3 stars = Neutral | 1-2 stars = Negative. 82% positive share creates severe class imbalance — accuracy is a misleading metric.

Appendix

3. Top Features using TF_IDF



Top predictive words:

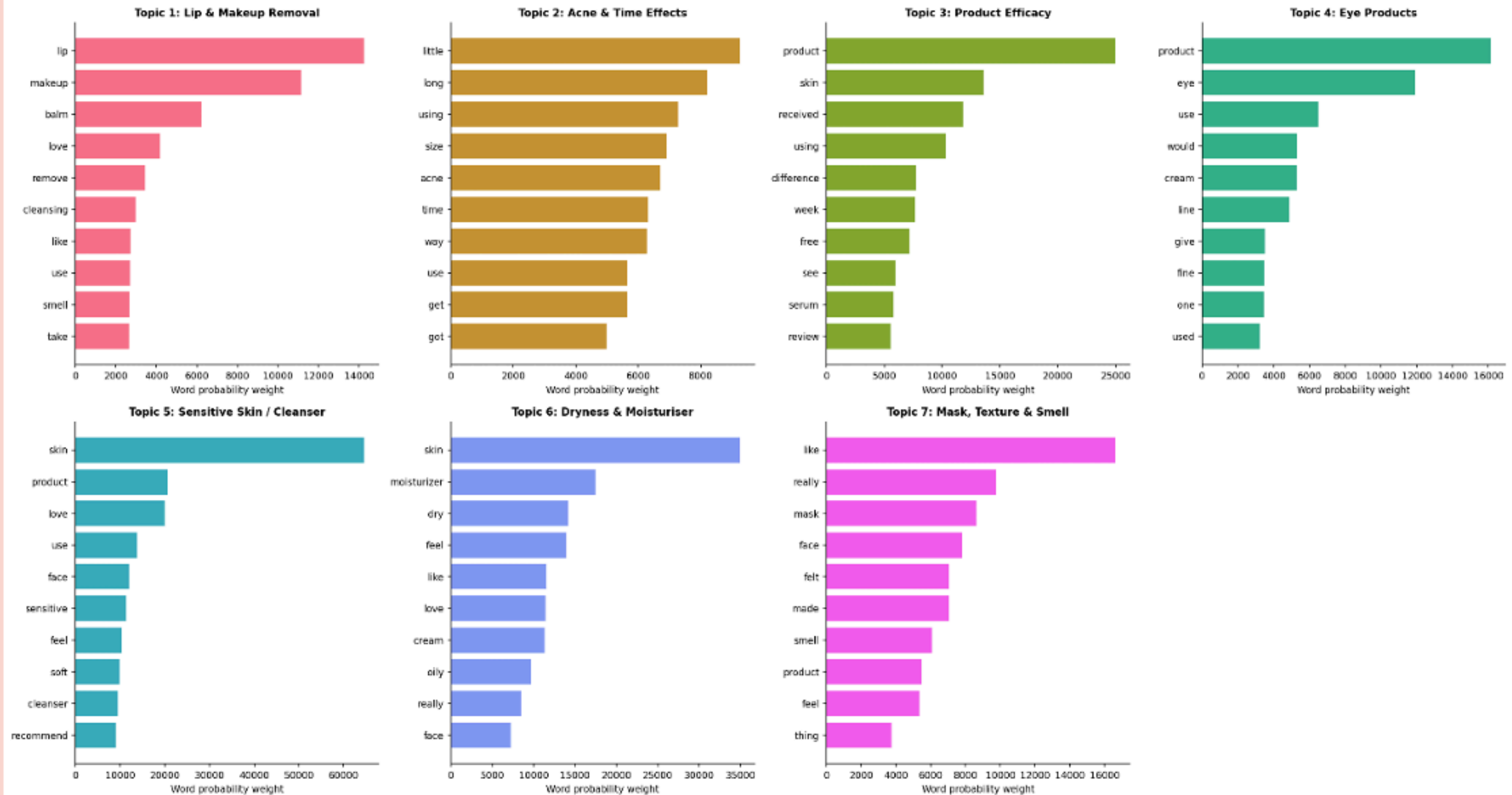
- Negative: 'disappointment', 'broke', 'returned'
- Positive: 'love', 'amazing', 'perfect'

LR is more explainable and interpretable, unlike BERT but they should be used in tandem

Appendix

4. LDA All Topics

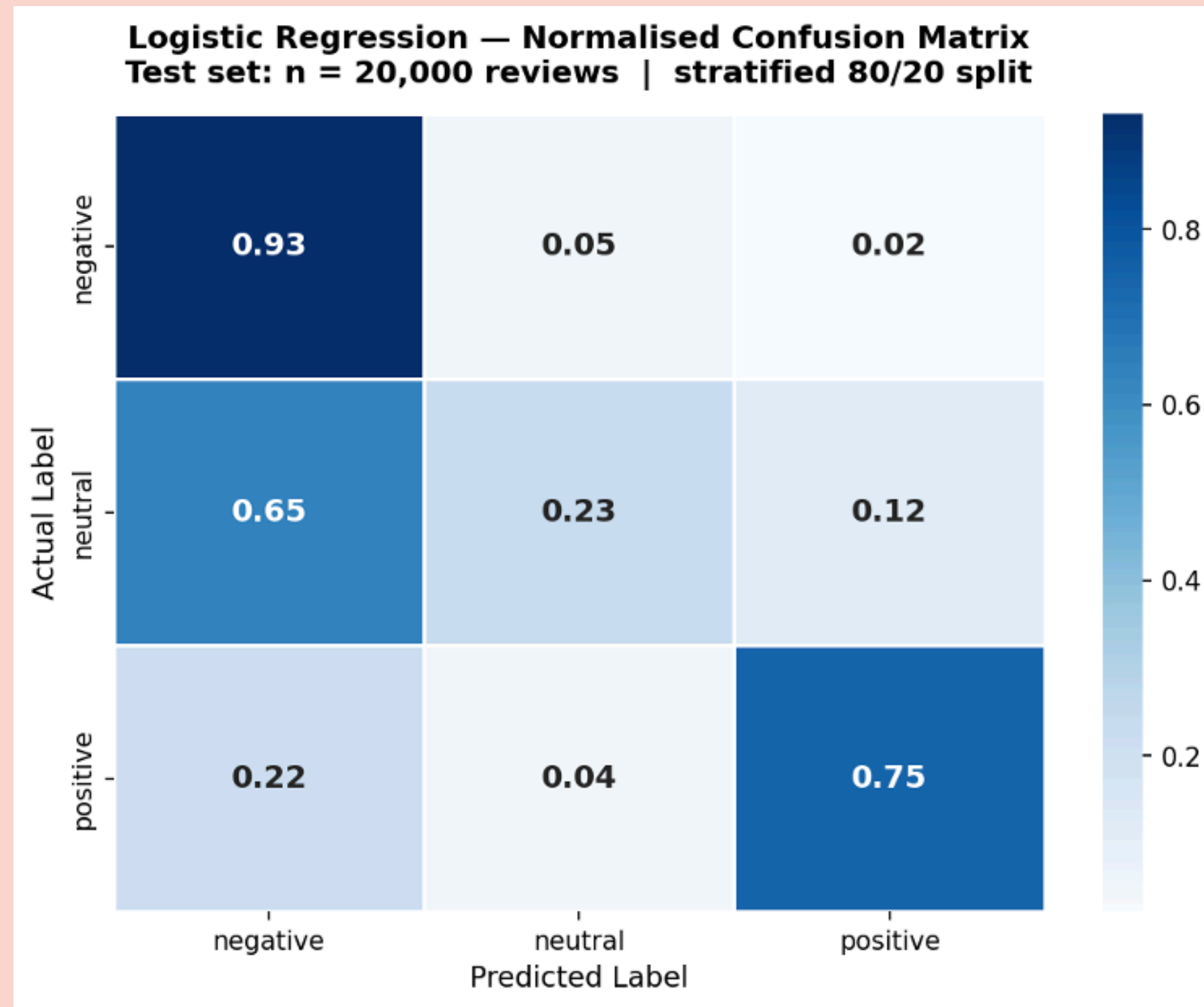
Fig LDA Topic Modelling — All Reviews (7 latent themes)



LDA on full 100k sample (all sentiment classes). Compare with Fig 11 (negative only) to identify complaint-specific vs general topics. General skincare discussion dominates when all classes are included.

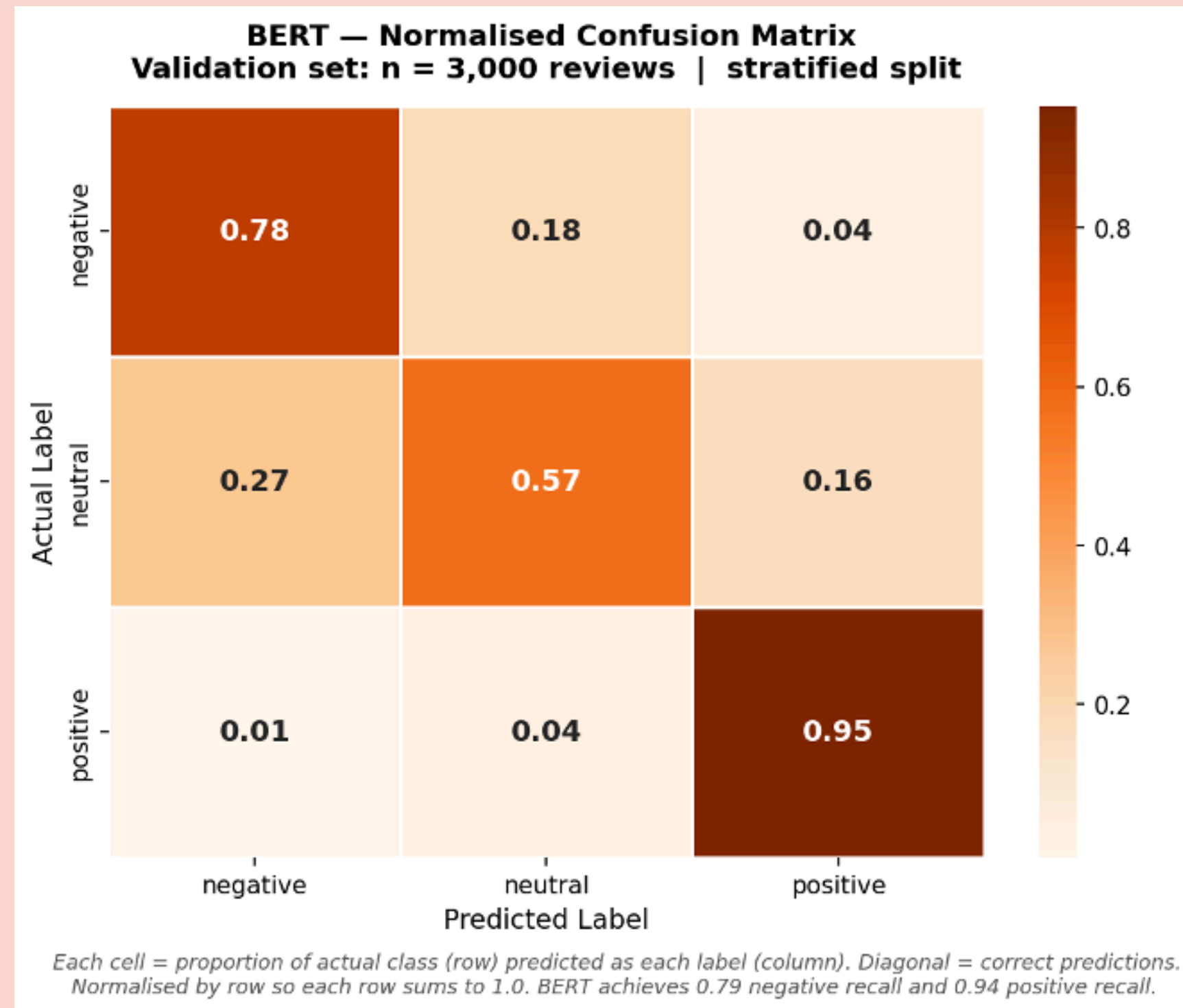
Appendix

5. LR Confusion Matrix



Appendix

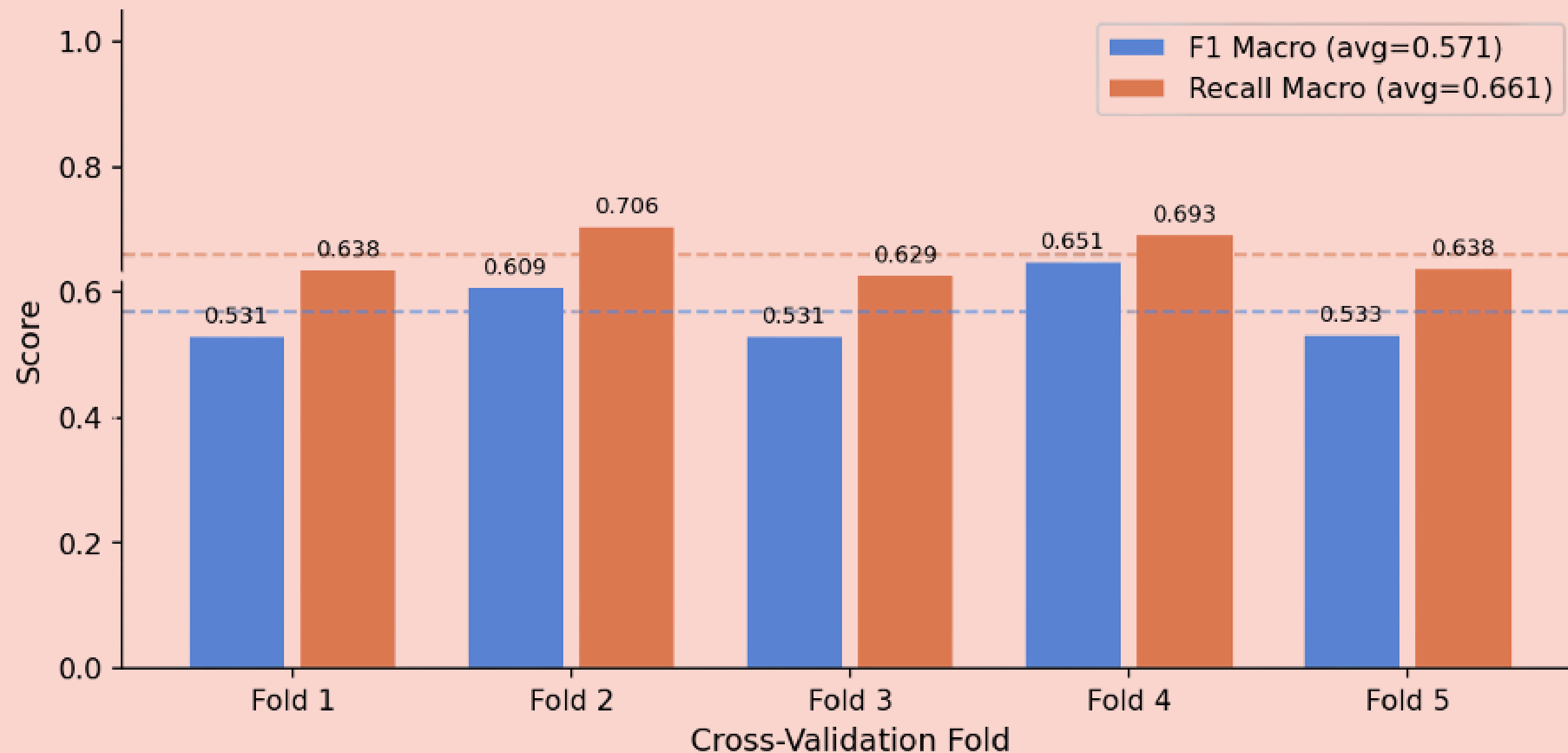
6. BERT Confusion Matrix



Appendix

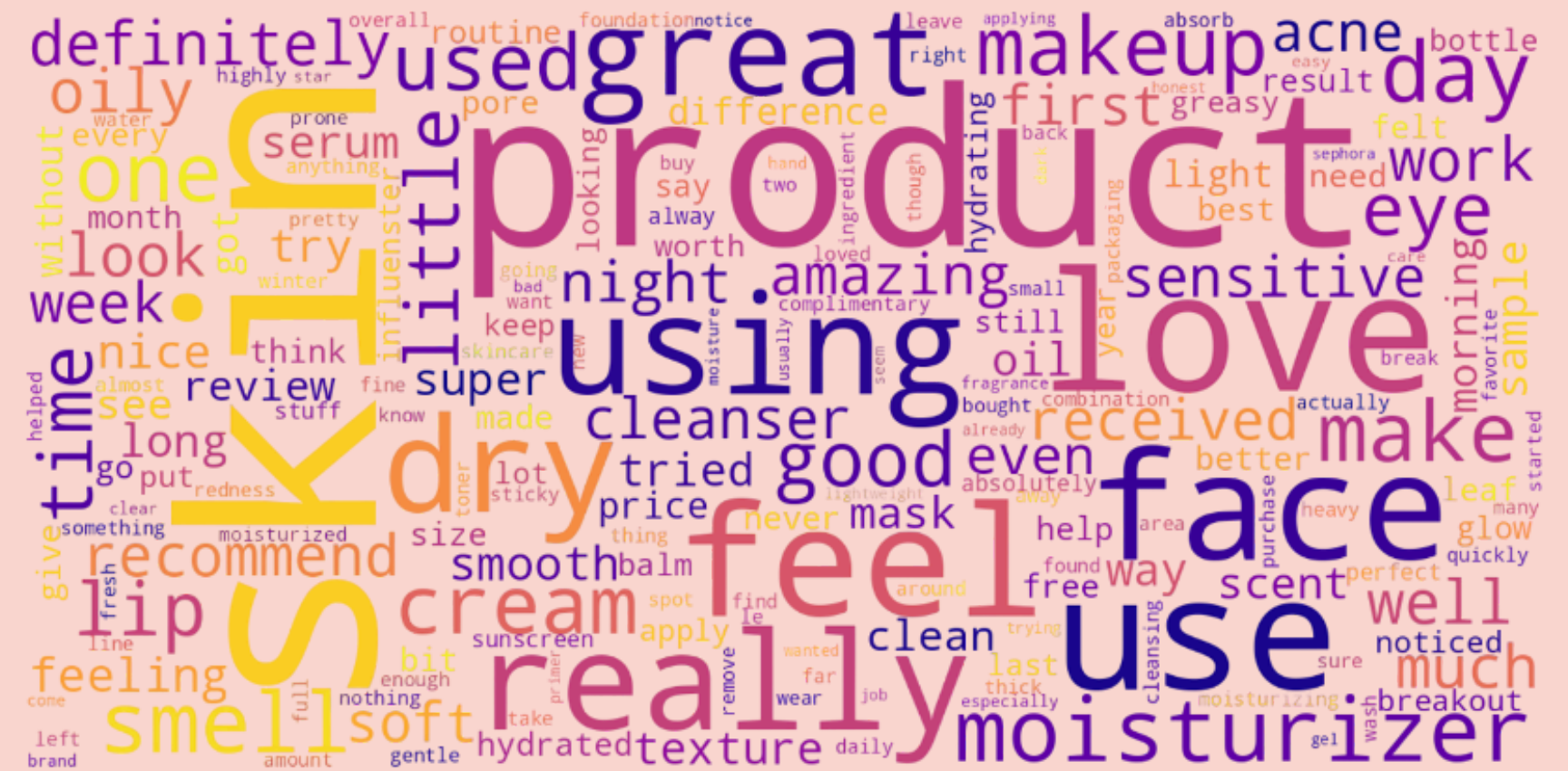
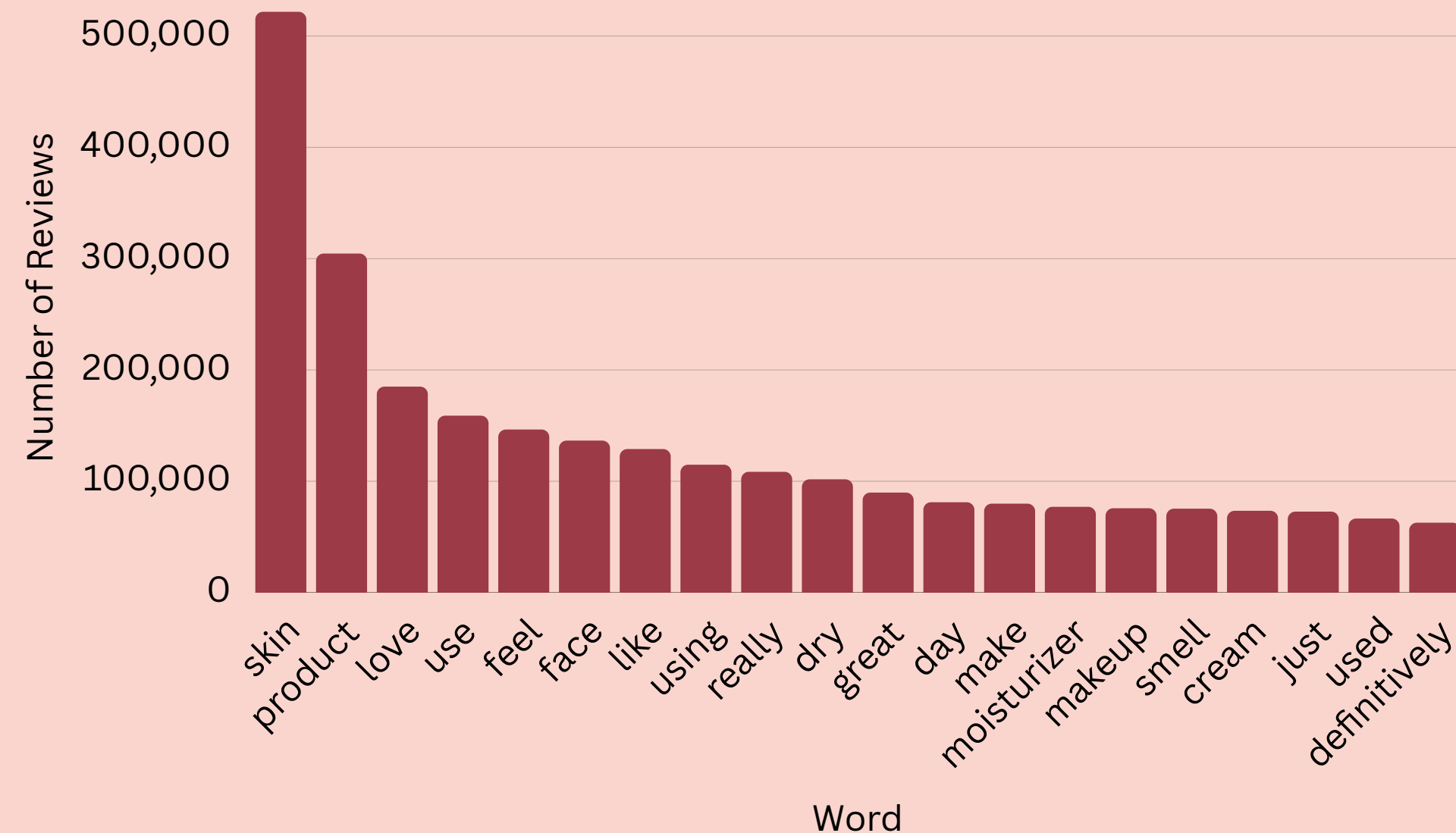
7. Cross Validation Results

5-Fold Stratified Cross-Validation — Logistic Regression
(Dashed lines = mean across folds)



8. Review Text Patterns: Positive Reviews

Top 20 Words in Positive Reviews



- Positive and negative reviews share much of the same topical vocabulary
- Stronger separation comes from context terms and phrases